

SURAJ GUPTA

ML Engineer · Software Developer

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[LinkedIn](#) | [GitHub](#)

Education

Delhi Technological University — B.Tech, Mathematics & Computing 2022 – 2026
CBSE Senior Secondary (PCM) | Percentage: 92.8% 2021
CBSE Secondary | Percentage: 94.0% 2019

Technical Skills

Languages: Python, C++, C, SQL, HTML, CSS
AI/ML & NLP: LLMs, LLaMA 3.3, LoRA Fine-Tuning, TF-IDF, Multinomial Naive Bayes, VADER Sentiment Analysis, Text Embeddings, Recommendation Systems, Scikit-learn, HuggingFace Transformers, SpaCy,
Frameworks & Tools: Flask, REST APIs, Git, GitHub, VS Code, Jupyter Notebook
CS Fundamentals: Data Structures & Algorithms, System Design, OOP, Operating Systems, DBMS, Computer Networks

Experience

Academic Mentor — Desh Ke Mentor Jun 2023 – Apr 2024

- Mentored 4 JEE/CBSE aspirants through structured bi-weekly sessions and personalised study plans, achieving consistent, measurable score improvements across all subjects. Tracked academic performance matrices and optimized learning plans.
- Designed concept-focused lesson modules and exam-pressure mock strategies with time-management frameworks, improving performance consistency and reducing revision time by ~25%. Result in improvement by average 20% of individual students.

Projects

MindEase — AI Mental Health Chatbot Jan 2026 – May 2026
Python · NLP · LLaMA 3.3 · Flask · REST API
[Github : Mindease-Chatbot](#) [Live Demo: MindEase](#)

- Architected a hybrid ML+LLM pipeline —TF-IDF vectorisation, Multinomial Naive Bayes intent classification (~85% accuracy), and VADER sentiment analysis —fed into a LLaMA 3.3 generative backend for context-aware, multi-turn mental health dialogue.
- Engineered real-time crisis detection using rule-based NLP triggers and sentiment thresholds; added Hinglish bilingual support via code-switching detection to serve 500+ Hindi-English bilingual users, enhancing inclusivity and conversational effectiveness.
- Deployed a production Flask REST API with session management, mood journaling, and wellness tracking— capable of handling 100+ concurrent sessions as a scalable full-stack web application. Implemented voice-enabled user interaction.

ATS Resume Analyzer May 2026 – Jun 2026
Python · Streamlit · Gemini API · NLP
[Github : ATS-Analyzer](#) [Live Demo: ATS-Analyzer](#)

- Built an AI-powered ATS Resume Analyzer that evaluates resume-job fit using semantic matching, keyword analysis, and NLP.
- Developed automated resume parsing and skill-gap detection to identify missing keywords and improve ATS compatibility.
- Integrated Google Gemini API to generate personalized resume reviews, intelligent rewrite suggestions, and tailored career learning roadmaps through an interactive Streamlit application, enhancing user engagement and career readiness.

AI Course Designer — LLM Recommendation Engine Sep 2025 – Dec 2025
Python · HuggingFace · SpaCy · LoRA Fine-Tuning

- Fine-tuned an LLM via LoRA (Low-Rank Adaptation) with structured prompt engineering to act as a course advisor, generating personalised Coursera recommendations — reducing irrelevant suggestions by 40% vs. baseline retrieval.
- Constructed an end-to-end NLP pipeline (SpaCy + HuggingFace Transformers) for skill extraction, normalisation, and vectorisation across 200+ courses, enabling precise skill-to-requirement mapping.
- Implemented a user feedback loop logging interaction history and preference signals to iteratively retrain the model, improving recommendation accuracy with each successive session. Integrated user preference analysis for more relevant learning pathways.

Research Publication

MindEase: A Hybrid NLP + LLM Architecture for Multilingual AI Mental Health Support *Accepted – May 2026*
International Scientific Journal of Engineering and Management (ISJEM) · Suraj Gupta

Presents design and evaluation of an NLP + LLM chatbot integrating sentiment analysis, crisis detection, and multilingual dialogue generation for real-world mental health support.

Positions of Responsibility

Logistics Head, EngiFest 2024 — Delhi Technological University 2024

- Directed end-to-end logistics for DTU's annual flagship fest with 250,000+ attendees, overseeing operational planning, resource allocation, and cross-functional coordination across multiple event domains.
- Managed and coordinated 120+ volunteers across 5 operational zones, ensuring seamless execution of artist hospitality, stage management, crowd operations, and merchandise distribution with zero operational delays.